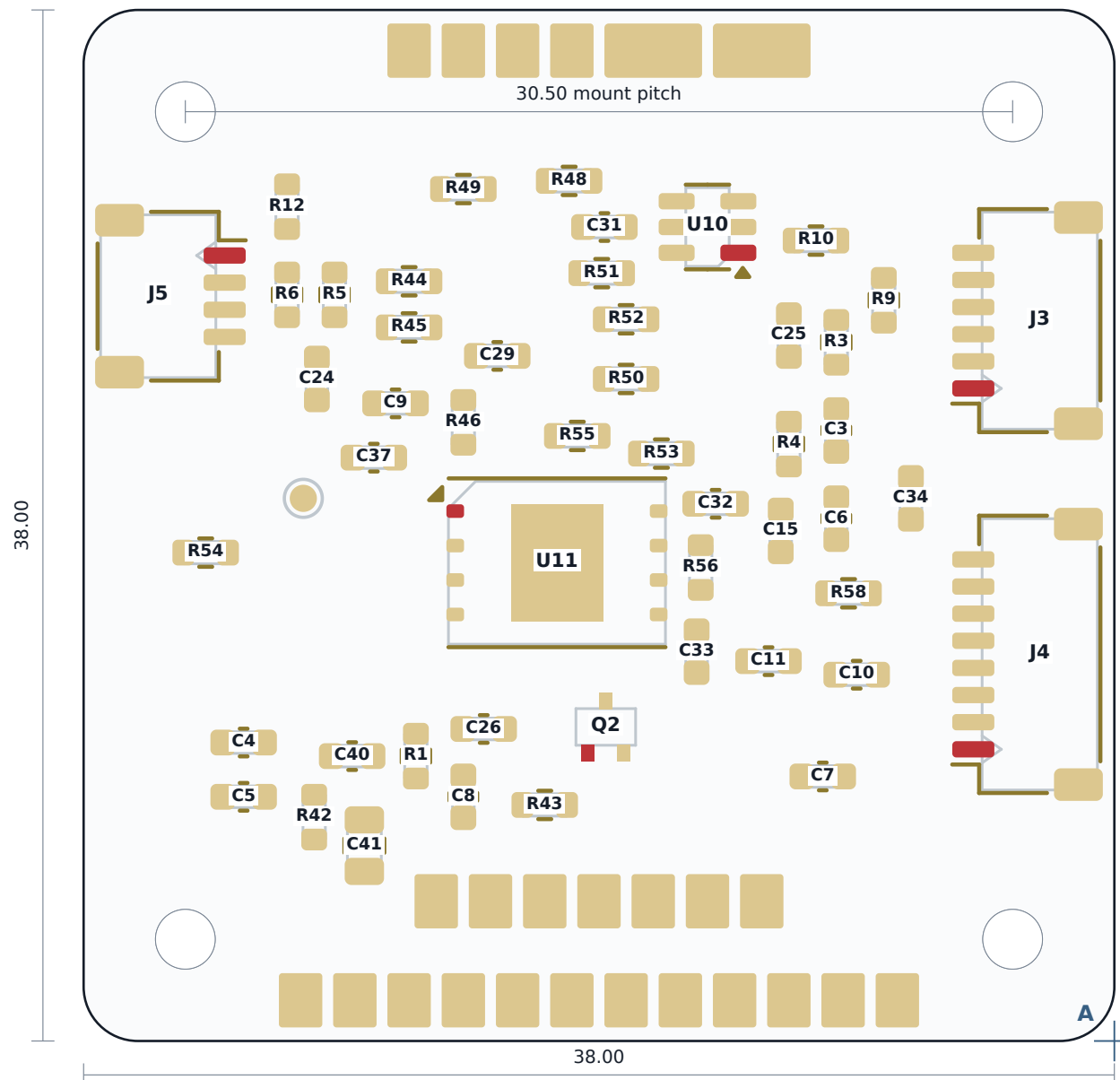


Component-side view | 4:1 at 100% print | Dimensions in mm | Component envelopes are native Fab geometry



Rear assembly view

Component-side view | 4:1 at 100% print | Dimensions in mm | Component envelopes are native Fab geometry



50 bottom-side fitted parts

Red lands identify pin 1, A1 or the diode cathode. Gold shows component lands; grey shows Fab outlines; ochre shows native silk. Reference text is an assembly overlay.

Rear is mirrored for human inspection

$X_{\text{view}} = 38 - X_{\text{native}}$

$Y_{\text{view}} = Y_{\text{native}}$

North stays at the top. Turn the board about a vertical axis to obtain this view.

Do not copy this mirror into CPL

The native audit uses $X = X_{\text{native}}$ and $Y = 38 - Y_{\text{native}}$ on both faces. Datum A therefore appears at the lower-right in this rear view.

U11 pin 1 is upper-left at native 180 degrees. Q2 Gate1 is lower-left at native 90 degrees; its custom footprint has no silk pin-1 cue. J3/J4 mouths face right here; J5 faces left.

Viewing limits

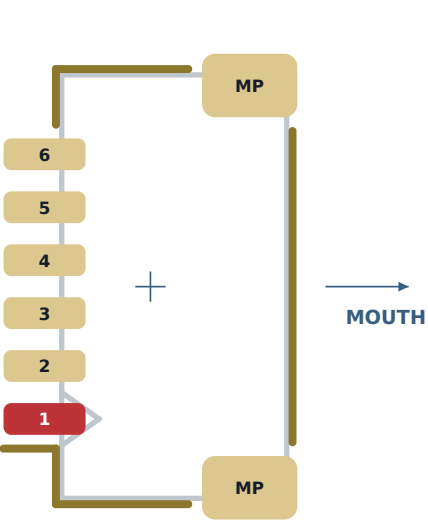
Routing is omitted from these assembly views. Stencil, fabrication, cable fit and supplier placement approval remain separate checks. ASSEMBLY NOT CLEARED: final placement, carrier, stencil and shell-solder production definitions are required. These views use native poses; page 6 records the supplier CPL corrections.

Side socket pin numbering

Rear component-side views | 7:1 at 100% print | Numbers are PCB pad numbers, not an unqualified mating-cable convention

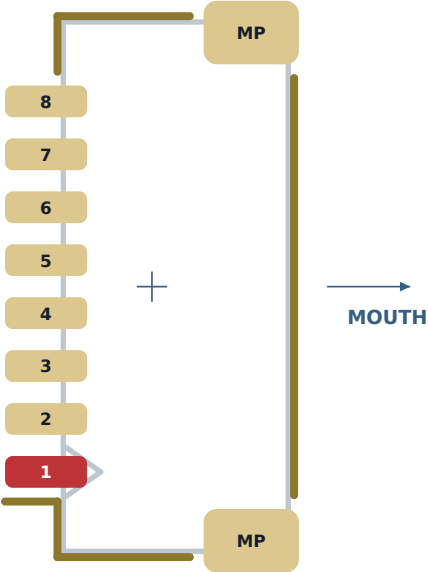
J3 MOTOR 6-wire

XDWF-0910-06P



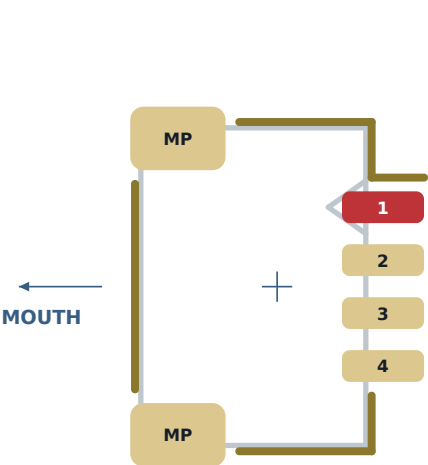
J4 External SPI sensor

SM08B-SRSS-TB(LF)(SN)



J5 GPS UART4

SM04B-SRSS-TB(LF)(SN)



Pad	Signal
1	+5V_STACK
2	GND
3	HOST_TX_LINK
4	MOTOR_TX_LINK
5	HOST_RUN_LINK
6	MOTOR_OK_LINK
MP	GND shell lands

Pad	Signal
1	3V3_CORE
2	GND
3	EXT_SPI_SCK
4	EXT_SPI_MOSI
5	EXT_SPI_MISO
6	EXT_SPI_CS
7	EXT_SENSOR_INT
8	GND
MP	GND shell lands

Pad	Signal
1	+5V_STACK
2	GND
3	UART4_TX
4	UART4_RX
MP	GND shell lands

Blue crosses mark native footprint anchors. Each socket body centre is offset 0.45 mm from its anchor. The native audit retains these anchors; the supplier CPL applies the user-provided corrections listed on page 6. J3 cavity numbering, housing/contact/wire compatibility and its fixture remain unqualified; the JST mounting-surface numbering evidence applies to J4/J5 only.

Placement datum and pin one reference

All coordinates below are native top-plan coordinates or the common CPL frame | Never use rear-view X as CPL X

Common file frame: native datum (0,38) mm; CPL X = native X and CPL Y = 38 - native Y. Bottom is side metadata; X is never negated. Angles below are native KiCad angles modulo 360. This table documents the native geometry; page 6 records the different user-supplied supplier angles and pickup offset. A supplier preview mark is not a universal guarantee of pin 1.

Ref	Exact manufacturer part	Side	Angle	Native anchor X,Y	Native audit CPL X,Y	Pin 1 / A1 native X,Y	Datum net
U1	STM32H743VIT6	Top	0	18, 20.5	18, 17.5	10.325, 14.5	NC (PE2)
U2	TPS62162DSGR	Top	0	5.05, 10.8	5.05, 27.2	4.1, 10.05	GND
U3	ICM-42688-P	Top	0	6.9, 29.6	6.9, 8.4	5.7375, 28.85	IMU_MISO
U4	USBLC6-2SC6	Top	180	30.5, 12.5	30.5, 25.5	31.6375, 13.45	USB_DM_CONN
U6	TPS22918DBVR	Top	0	32.1, 28.5	32.1, 9.5	30.9625, 27.55	USB_VBUS
U8	TPS3700DDCR	Top	0	23.3, 9.25	23.3, 28.75	22.1625, 8.3	STACK_5V_GOOD
U10	SN74LVC2G17DBVR	Bottom	0	15, 8	15, 30	13.8625, 8.95	MOTOR_TX_LINK
U11	W25N01GVZEIG	Bottom	180	20.55, 20.375	20.55, 17.625	24.3, 18.47	FLASH_CS
U12	TLV803EA29DBZR	Top	0	2.6, 17.8	2.6, 20.2	1.6625, 16.85	GND
U13	TPS2116DRLR	Top	90	28.5, 28.8	28.5, 9.2	27.75, 29.54	GND
Q1	2N7002	Top	90	34.65, 12.35	34.65, 25.65	33.7, 13.2875	USB_VBUS
Q2	DMN2310UW-7	Bottom	90	18.75, 26.425	18.75, 11.575	19.4, 27.375	USB_SOURCE_DISABLE
D4	ESDA7P120-1U1M	Top	0	35.6, 26.45	35.6, 11.55	34.975, 26.45	USB_VBUS
J2	TYPE-C-31-M-12	Top	90	34.35, 20	34.35, 18	30.305, 23.25	GND
J3	XDWF-0910-06P	Bottom	90	3.2, 11.45	3.2, 26.55	5.2, 13.95	+5V_STACK
J4	SM08B-SRSS-TB(LF)(SN)	Bottom	90	3.2, 23.75	3.2, 14.25	5.2, 27.25	3V3_CORE
J5	SM04B-SRSS-TB(LF)(SN)	Bottom	270	34.8, 10.55	34.8, 27.45	32.8, 9.05	+5V_STACK

Example: U11 bottom anchor is native (20.55,20.375), so CPL is (20.55,17.625). Its mirrored rear-view anchor appears at X_view = 17.45. That 17.45 is a drawing coordinate only. U8 has no silk pin-1 cue; use its Fab cue and this table. Q2 DMN2310UW-7 has no dedicated silk/Fab pin-1 marker: Gate1 is native (19.4,27.375), lower-left in the rear component view; Source2 is GND and Drain3 is USB_SWITCH_ON. Use the red lands and this table in supplier orientation review.

Six layer stack and assembly process

R3-S6-P6 | JLC06161H-3313 | ASSEMBLY NOT CLEARED: production definitions required

Layer / dielectric	Role or material	Thickness mm
L1 F.Cu	Signals / components	0.0350
L1-L2	3313 prepreg, Dk 4.1	0.0994
L2 In1.Cu	GND, front reference	0.0152
L2-L3	FR4 core, Dk 4.6	0.5500
L3 In2.Cu	Signals / power	0.0152
L3-L4	2116 prepreg, Dk 4.16	0.1088
L4 In3.Cu	GND, L3 reference	0.0152
L4-L5	FR4 core, Dk 4.6	0.5500
L5 In4.Cu	GND, rear reference	0.0152
L5-L6	3313 prepreg, Dk 4.1	0.0994
L6 B.Cu	Signals / components	0.0350

Thickness: order nominal 1.6 mm. Published finished thickness 1.54 mm +/-10%. Native copper/dielectric model totals 1.5384 mm; solder mask is unmodelled coating, not omitted.

Finish: 1 oz outer / 0.5 oz inner copper; green mask, white silk, ENIG.

USB basis: recorded JLC 90 ohm differential result W=0.1392 mm, gap=0.1600 mm, outer signal / adjacent GND, solder mask included. Final geometry and supplier controlled-impedance acceptance still apply.

Stencil and hidden joints

U2 uses the local DSG footprint: eight 0.50 x 0.25 mm lead apertures and two 0.90 x 0.70 mm EP windows, all R0.05; rounded EP coverage 87.332%. U11 has four 1.37 x 1.73 mm EP windows. Keep via drill holes out of these apertures and inspect hidden joints.

Production stencil definition

JLC regenerates SMT stencil data; require the real U2/U11 split windows in order remarks. Request evaluation of top 0.12 mm with a 0.10 mm U13 region, and bottom 0.10 mm. The stencil engineer must check step boundaries, nearby apertures and solder volume, or document a qualified alternative. This is a proposal, not process qualification.

USB shell soldering

J2 has two front 0.60 x 1.40 mm plated slots and two rear 0.60 x 1.70 mm slots, with 0.20 mm copper rings. No shell paste is provided. Require an accepted manual/selective or other qualified solder operation covering all four shell joints, then inspect them. Contacts-only reflow is insufficient. HRO specifies -30 to +80 C operation.

Panel and placements

Obtain the final supplier two-face 2D placement/polarity drawing for the exact 79-row corrected CPL. Use Standard double-sided PCBA with accepted carrier/rails/tooling/fiducials, connector mouth and depaneling access, with actual connector support/fixture suitability confirmed during order engineering. A customer-designed array is not required. Page 6 records the exact supplier-CPL corrections; the native placement audit remains separate.

Via fabrication settings

Select the 0.20 mm drill/small-via tier and epoxy-filled, copper-capped, mask-covered vias. Seven 0.35/0.20 mm vias have 0.075 mm radial annuli and meet the published preferred diameter-minus-drill rule; all other vias are at least 0.40 mm. Keep component slots and NPTH holes open. These are specified process selections.

Qualification boundary

Assembly production definitions are separate from bare-PCB fabrication. First-article checks cover the exact J3 harness and 1 A input-path budget, near-flush USB mating, HSE startup, rails, USB current/attachment firmware, thermal and EMI behaviour. CAD review does not establish these measured hardware results.

Primary process evidence: TI TPS6216x DSG | TI TPS2116 DRL | HRO USB drawing | JST SH | JLC assembly | JLC orientation | Stencil workflow | Via covering | J3 listing

Supplier CPL correction record

CPL corrections leave native lands and fitted poses unchanged | User file preserved | Residual assembly warnings remain

The user-provided placement file contains 79 unique references on the original sides. Its ten rotation changes and one USB pickup offset are reproduced below. A supplier rotation is a library-model convention; it does not rotate the native PCB lands. Native placement coordinates remain in the separate audit CSV.

Ref	Side	Native audit X,Y mm	Supplier X,Y mm	Native angle	Supplier angle
J2	Top	34.350000, 18.000000	33.000000, 18.000000	90.000000	90.000000
Q1	Top	34.650000, 25.650000	34.650000, 25.650000	90.000000	270.000000
Q2	Bottom	18.750000, 11.575000	18.750000, 11.575000	90.000000	0.000000
U1	Top	18.000000, 17.500000	18.000000, 17.500000	0.000000	270.000000
U4	Top	30.500000, 25.500000	30.500000, 25.500000	180.000000	90.000000
U6	Top	32.100000, 9.500000	32.100000, 9.500000	0.000000	180.000000
U8	Top	23.300000, 28.750000	23.300000, 28.750000	0.000000	270.000000
U10	Bottom	15.000000, 30.000000	15.000000, 30.000000	0.000000	90.000000
U11	Bottom	20.550000, 17.625000	20.550000, 17.625000	180.000000	0.000000
U12	Top	2.600000, 20.200000	2.600000, 20.200000	0.000000	180.000000
U13	Top	28.500000, 9.200000	28.500000, 9.200000	90.000000	0.000000

Source integrity: supplied CSV SHA-256 ed23023d67402462e10042c7a7e8467062def891a487c25afa7da5a2af2c00a2

Supplier rerun: The P6-Routing-Review reconciles current supplier findings against manufacturer lands and process evidence. J3 through-hole classification is a model error: the exact selected part and native lands have no drilled lead. Generic coloured rows do not replace the four production definitions on page 5. This page does not claim assembly clearance.

Required review: verify the final supplier two-face pin-1/polarity drawing, carrier and connector support, actual production stencil and all four USB shell solder joints before assembly. No purchase or assembly approval is implied by these files.